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Summary

This paper introduces a framework designed to enhance long-term, multi-turn collaboration between humans and large language models (LLMs). The core idea revolves around using an ensemble of Monte Carlo-based reward predictors to estimate the quality of LLM responses, coupled with a Bayesian meta-calibration technique to aggregate these predictions and quantify uncertainty. A key innovation is the integration of an uncertainty-

guided clarification mechanism, which dynamically triggers clarifying interactions when the model's confidence is low. The authors evaluate their approach across several tasks, including document editing, code generation, and mathematical problem-solving, reporting improvements in accuracy and ambiguity resolution compared to baseline methods. The methodology leverages ensemble learning to handle noisy reward signals, Bayesian methods to quantify uncertainty, and a dynamic clarification module to refine interactions. The empirical findings suggest that this framework can improve task performance and interaction efficiency in complex, multi-turn settings. The authors use synthetic datasets to test their approach, and they report improvements in metrics such as BLEU scores for document editing and accuracy in mathematical problem-solving. The paper's significance lies in its attempt to address the challenges of long-term human-LLM collaboration by incorporating uncertainty estimation and dynamic clarification, which are crucial for real-world applications. However, the paper also has several limitations, including a lack of clarity in certain aspects of the methodology, a reliance on synthetic data, and a lack of detailed analysis of computational costs and robustness. Despite these limitations, the paper presents a valuable contribution to the field of human-LLM interaction by proposing a novel framework that integrates several existing techniques in a new way. The paper's focus on uncertainty-guided clarification is particularly noteworthy, as it offers a practical approach to improving the reliability and effectiveness of LLMs in complex, multi-turn tasks. The authors' use of an ensemble of reward predictors and Bayesian meta-calibration is also a significant contribution, as it provides a robust mechanism for handling noisy reward signals and quantifying uncertainty. The paper's empirical findings, while limited to synthetic datasets, provide initial evidence that the proposed framework can improve task performance and interaction efficiency in various domains. Overall, the paper presents a promising approach to enhancing long-term human-LLM collaboration, but it also highlights the need for further research to address the identified limitations.

✓ Strengths

The paper's primary strength lies in its innovative integration of ensemble learning, Bayesian methods, and uncertainty-guided clarification to enhance long-term human-LLM collaboration. The use of an ensemble of Monte Carlo-based reward predictors is a well-designed approach for handling noisy reward signals, which is a common challenge in multi-turn interactions. The Bayesian meta-calibration technique effectively aggregates the predictions from the ensemble and provides a measure of uncertainty, which is crucial for triggering clarifications at the right time. The uncertainty-guided clarification module is a practical addition that dynamically improves interaction quality by reducing ambiguities, particularly useful in complex, multi-turn tasks. The paper also demonstrates promising improvements in task-specific metrics, such as BLEU scores for document editing and accuracy in mathematical problem-solving, suggesting its potential for real-world applications. The authors' approach to combining these techniques is a novel contribution, even if the individual components are not entirely new. The paper's focus on long-term collaboration is also a significant strength, as it addresses a critical challenge in the field of human-LLM interaction. The empirical results, while limited to synthetic datasets, provide initial evidence that the proposed framework can improve task performance and interaction efficiency across various domains. The ablation studies, which analyze the impact of different window sizes and Monte Carlo sample counts, further support the robustness of the approach. The paper's mathematical grounding, while not always presented with complete clarity, provides a solid foundation for the proposed framework. The authors' attempt to address the challenges of long-term human-LLM collaboration by incorporating uncertainty estimation and dynamic clarification is a significant contribution to the field. The paper's focus on practical applications, such as document editing and code generation, makes it relevant to real-world scenarios. The use of an ensemble of reward predictors and Bayesian meta-calibration is a significant technical innovation, as it provides a robust mechanism for handling noisy reward signals and quantifying uncertainty. The paper's empirical findings, while limited to synthetic datasets, provide initial evidence that the proposed framework can improve task performance and interaction efficiency in various domains. Overall, the paper presents a promising approach to enhancing long-term human-LLM collaboration, and its strengths lie in the novel integration of existing techniques and the focus on uncertainty-guided clarification.

✗ Weaknesses

Upon careful examination, several weaknesses in this paper significantly impact its overall contribution. First, the paper suffers from a lack of clarity in its methodology, particularly in the definition of key terms and the implementation of core components. For instance, the term "interactivity score" is mentioned as part of the intrinsic reward $R_{int}(t)$ in Section 2, but it is not explicitly present in the provided formula, creating confusion about its precise nature and calculation. This lack of clarity extends to the description of the Bayesian meta-calibrator, where the paper mentions using Bayesian linear regression but fails to specify the prior distributions, likelihood functions, or inference algorithms used. Similarly, the uncertainty-guided clarification module is described at a high level, without detailing how clarification questions are generated or incorporated into the conversation. This lack of detail makes it difficult to fully understand and reproduce the proposed method. My confidence in this weakness is high, as

it is directly evident from the paper's text and the absence of crucial details. Second, the paper's methodological contribution is limited by its reliance on combining existing techniques without introducing substantial innovation. While the integration of ensemble learning, Bayesian methods, and uncertainty-guided clarification is a reasonable approach, the paper does not present a novel theoretical framework or a significant advancement in any of these individual areas. The core idea of using uncertainty to guide clarification is not new, and the paper does not adequately demonstrate how their specific implementation offers a substantial improvement over existing methods. The paper acknowledges existing work in these areas, but it does not clearly articulate the novelty of its approach beyond the specific combination of these techniques. My confidence in this weakness is medium, as the paper does attempt to integrate existing techniques in a novel way, but the lack of substantial innovation is evident. Third, the experimental evaluation relies heavily on synthetic data, raising concerns about the real-world applicability of the proposed method. The paper uses synthetic datasets for four distinct domains, but it lacks a thorough evaluation on real-world datasets, making it difficult to assess the practical effectiveness of the proposed method. Furthermore, the paper does not provide sufficient details about the synthetic data generation process, making it hard to evaluate the validity of the experimental results. This reliance on synthetic data limits the generalizability of the findings and raises questions about the method's performance in real-world scenarios. My confidence in this weakness is high, as the paper explicitly states the use of synthetic data and lacks details on its generation. Fourth, the paper lacks a detailed analysis of the computational overhead introduced by the ensemble and Bayesian methods. The paper describes the use of Monte Carlo sampling and ensemble methods, but it does not provide any analysis of the computational costs or time complexity of these components. This lack of analysis makes it difficult to assess the practical feasibility of the approach, particularly for resource-constrained environments. The paper also does not quantify the memory requirements of the ensemble, which is an important consideration for practical applications. My confidence in this weakness is high, as the paper does not include any discussion or data on computational costs. Fifth, the paper's clarification mechanism uses a fixed threshold for triggering clarifications, which may not be optimal across different tasks or user interactions. The paper does not explore adaptive strategies for triggering clarifications based on user preferences or task complexity. This lack of adaptability may limit the system's overall efficiency and user experience. My confidence in this weakness is high, as the paper explicitly states the use of a fixed threshold. Sixth, the paper lacks a thorough discussion of potential biases in the reward models and how these biases might affect the system's performance across different user groups or tasks. The paper does not analyze the sensitivity of the system to biased reward signals, which is a crucial consideration for real-world applications. My confidence in this weakness is high, as the paper does not include any discussion regarding potential biases. Finally, the paper's performance in tasks outside the tested domains, such as creative writing or open-ended dialogues, remains unexplored. The current evaluation focuses on tasks with relatively well-defined metrics, and it is unclear how the framework would perform in more open-ended tasks where the notion of success is less clear. My confidence in this weakness is high, as the paper explicitly lists the tested domains and lacks results for other types of tasks. These weaknesses, taken together, significantly limit the paper's overall contribution and highlight the need for further research to address these issues.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors must significantly enhance the clarity of their methodology. This includes providing precise definitions for all notations and terms used in the paper, including $R_{ext}(t,g)$, $R_{int}(t)$, and the interactivity score. The formulation of the intrinsic reward should be explained step-by-step, with a clear rationale for each component. The authors should also provide a more detailed explanation of the Monte Carlo-based reward predictors, including the specific algorithms used and how the samples are generated. Furthermore, the paper should include a more thorough discussion of the limitations of using synthetic data and how these limitations might affect the generalizability of the results. The authors should consider including a small-scale evaluation on real-world data to demonstrate the practical applicability of their method. Second, the authors should strengthen the methodological contribution of the paper by clearly articulating the novelty of their approach and how it differs from existing methods. Instead of simply combining existing techniques, the authors should focus on introducing a novel aspect or a significant improvement to the existing methods. For example, they could explore a new way of combining the ensemble predictions, or they could develop a more sophisticated uncertainty-guided clarification mechanism. The paper should also include a more detailed analysis of the theoretical properties of the proposed method, such as convergence guarantees or bounds on the estimation error. The authors should also provide a more thorough comparison with existing methods, highlighting the advantages and disadvantages of their approach. Third, the authors should provide more implementation details, particularly regarding the Bayesian meta-calibrator and the uncertainty-guided clarification module. The authors should specify the exact type of Bayesian linear regression used, including the prior distribution and the likelihood function. They should also explain how the uncertainty estimates are derived from the ensemble of predictors and how these estimates are used to trigger the clarification module. The paper should also provide details on how the clarification questions are generated and incorporated into the conversation, including the specific algorithms and techniques used. The authors should consider releasing their code to allow for reproducibility and further analysis by the research community. Fourth, the authors should address the computational cost concerns by providing a detailed analysis of the time complexity for each component of the framework, including the ensemble reward estimation, Bayesian uncertainty quantification, and clarification mechanisms. This analysis should include both theoretical estimates and empirical measurements on different hardware configurations. Furthermore, the authors could explore techniques to optimize the computational efficiency of the framework, such as using more efficient sampling

methods or reducing the number of ensemble members. For example, investigating the trade-off between the number of Monte Carlo samples and the accuracy of the uncertainty estimates could lead to significant performance improvements. Additionally, the authors could consider using approximation methods for Bayesian inference to reduce the computational overhead. Fifth, the authors should improve the clarification mechanism by exploring adaptive strategies for triggering clarifications based on user preferences and task complexity. This could involve incorporating user feedback on the frequency and usefulness of clarifications, as well as using task-specific characteristics to adjust the clarification threshold. For example, in tasks requiring high precision, the system could be configured to ask for clarification more frequently, while in tasks where speed is crucial, the threshold could be raised to reduce the number of interruptions. The authors could also investigate the use of reinforcement learning to optimize the clarification strategy based on the observed outcomes. Sixth, the authors should conduct a more thorough analysis of potential biases in the reward models and their impact on the system's performance. This analysis should include evaluating the reward model's performance across different demographic groups or task types to identify any potential biases. The authors could also explore techniques to mitigate these biases, such as using adversarial training or re-weighting the training data. Finally, the authors should investigate the framework's performance in more open-ended tasks, such as creative writing or open-ended dialogues, to assess its generalizability. This would involve defining appropriate evaluation metrics for these tasks and conducting experiments to evaluate the framework's performance. A more comprehensive evaluation across diverse tasks and user groups would greatly strengthen the paper's claims and demonstrate the robustness of the proposed approach.

 Questions

Based on my analysis, several key questions remain unanswered. First, what is the precise formulation of the intrinsic reward $R_{int}(t)$? The paper mentions an "interactivity score" without defining it. How is this score calculated, and what is the rationale behind including it in the intrinsic reward? This is crucial for understanding the reward mechanism and its impact on the system's behavior. Second, how does the proposed method compare to other state-of-the-art techniques for long-term reward estimation in human-LLM collaboration? The paper could benefit from a more detailed comparison with existing methods, highlighting the advantages and disadvantages of the proposed approach. This would help to contextualize the paper's contribution and identify areas for improvement. Third, what language model is used in the experiments? The paper does not specify the exact model used, which makes it difficult to reproduce the results. This information is essential for ensuring the reproducibility and validity of the findings. Fourth, given that the LLM is considered "large," is the input length a factor in the method's performance? The paper does not discuss the impact of input length on the method's effectiveness. This is an important consideration for practical applications, as long input sequences may pose challenges for the proposed method. Fifth, what does "active fine-tuning" refer to in the context of this paper? The term is used without a clear explanation. Understanding the specifics of the fine-tuning process is crucial for evaluating the paper's methodology. Sixth, how is the clarification efficiency metric calculated? The paper mentions this metric but does not provide a clear definition or formula. A clear definition of this metric is necessary for understanding the paper's evaluation of the clarification mechanism. Seventh, how does the framework handle scenarios where the LLM's responses are only partially ambiguous? Is there a threshold for uncertainty that triggers clarification, and how is this threshold determined? This is important for understanding the system's behavior in complex, real-world scenarios. Eighth, could the authors elaborate on the computational overhead introduced by the ensemble and Bayesian methods? How does the framework perform in real-time applications with limited computational resources? This is a critical consideration for the practical deployment of the proposed method. Ninth, how does the framework address potential biases in the reward models, especially in diverse user groups or tasks? This is an important consideration for ensuring the fairness and reliability of the system. Finally, how well does the framework generalize to tasks outside the tested domains, such as creative writing or open-ended dialogues? This is crucial for assessing the broader applicability of the proposed method. These questions highlight key uncertainties and areas where further clarification is needed to fully understand and evaluate the paper's contribution.

 Scores

Soundness: 2.75

Presentation: 2.0

Contribution: 2.75

Rating: 5.25

 AI Review from ZGCA will be automatically processed

Summary

The paper proposes a framework for multi-turn human-LLM collaboration that (i) estimates a conversation-level reward $R^*(t|g) = R_{\text{ext}}(t,g) + R_{\text{int}}(t)$, where R_{ext} is task-specific (e.g., BLEU, unit test pass rate, accuracy) and R_{int} includes a token-length penalty and an LLM-based interactivity score; (ii) uses an ensemble of Monte Carlo reward predictors under different window sizes and sample counts; (iii) aggregates ensemble predictions via Bayesian linear regression ("Bayesian meta-calibration") and computes an uncertainty metric (sample standard deviation of ensemble predictions); and (iv) triggers a clarification mechanism when uncertainty exceeds a threshold $\tau=0.15$. The method reportedly improves MATH accuracy (0.739 $\rightarrow$ 0.799) and ambiguity resolution (0.8 $\rightarrow$ 1.0) with smaller gains on document editing BLEU (0.625 $\rightarrow$ 0.637), but performance decreases for code generation (0.532 $\rightarrow$ 0.489). Experiments are conducted on synthetic datasets across four domains, with hyperparameters $\lambda=0.01$ (token penalty) and a fixed bonus $\delta=0.05$ added when clarification is triggered.



Strengths

- Timely problem: optimizing long-horizon, multi-turn human-LLM collaboration rather than single-turn quality.
- Conceptual integration of ensemble uncertainty with a clarification trigger is intuitive and potentially impactful for interactive systems.
- Conversation-level reward decomposition that explicitly balances extrinsic outcomes with interaction efficiency is a reasonable modeling choice.
- Empirical indications of gains in certain domains (MATH-Chat, Abg-CoQA) and reduced token usage in document editing.



Weaknesses

- Insufficient empirical validation: only 5 candidate responses per domain are used (Section 5), which is far too small to support claims of statistical significance or robustness for inherently variable LLM outputs.
- Key methodological gaps: $R_{\text{LLM}}(t)$ (the interactivity score) is never defined operationally; the precise training/targets for the Bayesian linear regression aggregator are not specified; the uncertainty metric is simply the sample standard deviation of ensemble predictions without a coherent Bayesian treatment tied to the regression posterior.
- The clarification mechanism is unclear and potentially confounded: Methods state that when $\sigma_{\text{agg}} > \tau$, a fixed bonus $\delta=0.05$ is added to the reward (Equation for $\hat{R}_{\text{adj}}$), which functions as reward shaping rather than evidence of a real clarifying dialogue round that changes the model's output; the paper does not describe how clarification prompts are generated, how many additional turns occur, or how those turns alter outputs.

- Claims of “extensive ablation studies” are not supported by reported results; no ablation tables/figures are shown despite assertions in the abstract and Section 6.
- Baselines are under-specified: “without clarification triggers” is too vague; there are no architectural or training details for the base LLM, LoRA setup, or reward predictors, hindering reproducibility and fair comparison.
- Synthetic datasets and narrow evaluation (with tiny sample sizes, no confidence intervals, and no significance testing) limit external validity; code generation performance degrades, indicating domain fragility.
- Some modeling assumptions (e.g., independence of Monte Carlo samples, stationarity of token cost) are strong and not empirically probed; τ and δ appear tuned without principled justification.

? Questions

- Please define $R_LLM(t)$ precisely: what is the interactivity score, how is it computed, calibrated, and validated? Is it predicted by a learned model or derived from rules?
- What exactly is the Bayesian linear regression used for meta-calibration? What are the inputs (ensemble predictions), outputs (ground-truth targets), priors, and training procedure? If there is no ground-truth reward during aggregation, what is the regression target? How do you obtain labels for calibration?
- Uncertainty: If BLR is used, why is uncertainty reported as the sample standard deviation of ensemble predictions instead of the posterior predictive variance from the regression model? Please reconcile this with the claimed Bayesian methodology.
- Clarification mechanism: Do you actually conduct additional clarification turns with the user or an LLM agent that modifies the candidate response? If so, provide the prompting protocol, number of extra turns, and how the final answer changes. If not, how does adding a fixed bonus δ constitute a clarification?
- Ablations: You claim extensive ablations on window size w and sample count S . Please provide the ablation tables with metrics, confidence intervals, and compute costs. Which configurations materially contribute to gains?
- Statistical rigor: With only 5 candidate responses per domain, how do you establish statistical significance? Please provide confidence intervals or nonparametric tests over substantially larger samples (e.g., 100+ prompts per domain), including multiple seeds.
- Baselines and training: Describe the base LLM(s), LoRA details, training data sizes, optimization settings, and how many steps are run. Are the baselines trained/fine-tuned identically except for the clarification mechanism?
- Thresholds and hyperparameters: How were $\tau=0.15$ and $\delta=0.05$ chosen? Is there validation showing sensitivity curves for τ and δ and domain-specific tuning?
- External validity: Are the datasets public and realistic? For document editing, why BLEU vs task-specific human preference or edit-distance measures? For code, what unit tests and harness were used?
- Compute and overhead: Report the computation cost of ensembles, the clarification frequency, and wall-clock/throughput impacts.

! Limitations

- The token-based penalty may incentivize brevity at the expense of necessary detail, potentially hurting accessibility and safety-critical explanations.
- Clarification triggers can increase cognitive load or interaction burden for users; user-centric metrics (satisfaction, burden) are not evaluated.
- The framework assumes stationary token costs and independent Monte Carlo samples, which may not hold in practice and can bias uncertainty estimates.
- Synthetic datasets and limited samples constrain generalizability; domain-specific failures (code generation) suggest the approach may need careful adaptation.

- If real user data were employed, there would be privacy considerations around storing and analyzing clarification dialogues; guidance on data handling is absent.
- Ensemble methods add computational and energy overheads that may hinder deployment in resource-limited settings.

 Image Evaluation

Cross-Modal Consistency: 24/50

Textual Logical Soundness: 17/30

Visual Aesthetics & Clarity: 9/20

Overall Score: 50/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

- Major 1: The uncertainty trigger is central but Fig. 1 shows no τ line or trigger markers, so the mechanism cannot be verified. Evidence: Sec. 4: “Clarification Trigger = 1, if $\sigma_{agg} > \tau$ ($\tau=0.15$).”
- Major 2: Claimed baseline → active gains are not visually supported; Fig. 2 shows single-mode bars and mixes units, preventing delta verification. Evidence: Sec. 6: “baseline 0.739 to 0.799”; Fig. 2 caption “Evaluation Metrics Across Domains”.
- Major 3: Dataset naming inconsistency (“BigCodeBench-Chat” vs “BigIntCodeBench-Chat”) confuses mapping between text, tables, and figures. Evidence: Sec. 5 “BigIntCodeBench-Chat”; Sec. 1/6 “BigCodeBench-Chat”.
- Major 4: Table numbering inconsistent; “Table 6” is referenced but multiple “Table 1” also appear, and numbering is unclear. Evidence: Sec. 6: “A detailed quantitative comparison is presented in Table 6.”
- Minor 1: Aggregation described as Bayesian linear regression, yet only a sample std-dev is reported; no posterior variance or weights shown. Evidence: Sec. 4: “ $\sigma_{agg} = \sqrt{(1/N \sum (r_i - \hat{R})^2)}$.”
- Minor 2: Figure 2 axis label “Metric Score” is unitless despite mixing accuracy and token count.

2. Text Logic

- Major 1: The interactivity term $RLLM(t)$ is invoked but never defined/measured, leaving R_{int} partly unspecified. Evidence: Sec. 1: “augmented by an LLM-based interactivity score.”
- Minor 1: Fixed bonus $\delta=0.05$ adds to reward regardless of actual outcome; justification and calibration are unclear. Evidence: Sec. 4: “ $\hat{R}_{adj} = \hat{R} + \delta$, $\delta = 0.05$.”
- Minor 2: Statistical significance claims are weak given very small n. Evidence: Sec. 5: “each experiment is run on 5 candidate responses.”

3. Figure Quality

- Major 1: Fig. 2 mixes incommensurate scales (accuracy/F1 vs token count) on one y-axis; relative heights are misleading. Evidence: Fig. 2: orange bar ≈ 50 vs accuracies $\approx 0-1$.
- Major 2: Fig. 1 lacks τ line, legend, or markers for “clarification triggered” vs not; “Active Mode” alone is insufficient to convey the intended message. Evidence: Fig. 1: scatter only; no annotations.
- Minor 1: Small tick labels and absent value annotations hinder quick reading.

Key strengths:

- Clear motivation and compact mathematical specification of reward and trigger.

- Broad, multi-domain evaluation intent.
- Key weaknesses:
- Core mechanism (uncertainty-triggered clarifications) not demonstrable from figures.
 - Inconsistent dataset naming and table numbering.
 - Undefined RLLM term; ablation claims lack concrete visual/tabular evidence.
 - Fig. 2's mixed units and single-mode plotting obscure claimed improvements.

- Recommendations:
- Add τ vertical line, clarify-trigger markers, and before/after coloring in Fig. 1.
 - Split Fig. 2 into separate axes or panels; include baseline vs active with exact values.
 - Unify dataset names and fix table numbering.
 - Define and operationalize RLLM(t); provide ablation tables/plots with w and S.

 Scores

Originality: 3

Quality: 2

Clarity: 2

Significance: 2

Soundness: 2

Presentation: 2

Contribution: 2

Rating: 3

 AI Review from SafeReviewer will be automatically processed



Summary

This paper introduces a framework aimed at enhancing long-term, multi-turn collaboration between humans and large language models (LLMs). The core idea revolves around optimizing conversation-level rewards by combining task-specific success metrics with penalties for inefficiency, particularly regarding token usage. The authors propose an ensemble of Monte Carlo-based reward predictors, which are then aggregated using Bayesian linear regression to provide both a reward estimate and an uncertainty metric. This uncertainty metric is crucial, as it triggers an uncertainty-guided clarification module when a predefined threshold is exceeded. This module dynamically introduces clarification rounds to refine the LLM's responses. The authors evaluate their approach across four tasks: document editing, code generation, mathematical problem-solving, and ambiguity resolution. They report improvements in metrics such as BLEU score for document editing and accuracy for mathematical problem-solving. However, the results are mixed, with a decrease in unit test pass rates for code generation. The paper emphasizes the importance of balancing immediate response quality with long-term conversational effectiveness, and the proposed framework is presented as a step towards achieving this goal. The methodology involves a two-phase training pipeline, including pretraining on synthetic dialogues and active fine-tuning. The authors also conduct ablation studies to analyze the impact of different window sizes and Monte Carlo sample counts. Overall, the paper presents an interesting approach to multi-turn human-LLM collaboration, but it suffers from several limitations in its presentation and evaluation.

For instance, the paper's introduction immediately dives into technical details, using equations and specific parameters without first establishing the broader context of the problem. Terms like 'ensemble-based reward estimation,' 'Bayesian meta-calibration,' and 'Monte Carlo sample counts' are introduced without sufficient explanation, making it difficult for a general audience to grasp the core concepts. The 'main idea' section also starts with equations and technical jargon, assuming prior knowledge of multi-turn reward estimation and Bayesian methods. The term 'Bayesian meta-calibration' is used without a clear definition, and the explanation of how uncertainty is quantified is vague. The 'method' section lacks a clear, high-level overview, jumping directly into technical details without explaining the overall workflow or the interaction between the ensemble predictors, Bayesian meta-calibrator, and the clarification module. The 'experiments' section presents results without sufficient context, making it hard to understand the significance of the reported metrics. The 'results' section also suffers from a lack of detailed analysis, failing to explain the reasons behind the observed trends. The paper also lacks a clear explanation of how the synthetic data is generated and validated, and the experimental setup lacks crucial details, such as the specific LLMs used, the exact implementation of the ensemble predictors, and the hyperparameter tuning process. The paper also does not include a direct comparison with CollabLLM, a method mentioned in the related work, and the ablation studies do not provide sufficient insights into the contribution of individual components. The paper also does not discuss the computational overhead of the proposed method, which is an important consideration for practical applications. Finally, the paper lacks a dedicated 'Limitations' section, and the discussion of future work is brief and lacks specific details.



Strengths

Despite the identified weaknesses, the paper does present some notable strengths. The core idea of using an ensemble of Monte Carlo-based reward predictors combined with Bayesian meta-calibration to quantify uncertainty is a promising approach. This method provides a principled way to estimate both the reward and its associated uncertainty, which is crucial for triggering clarifying interactions. The use of an uncertainty-guided clarification module is also a valuable contribution, as it allows the system to dynamically adapt to ambiguous situations and improve the quality of its responses. The paper's attempt to balance task-specific success with efficiency, particularly through the token-based cost term, is also a positive aspect. The authors also conduct ablation studies to analyze the impact of different window sizes and Monte Carlo sample counts, which provides some insights into the robustness of the method. The paper's evaluation across four diverse tasks—document editing, code generation, mathematical problem-solving, and ambiguity resolution—demonstrates the potential applicability of the proposed framework in different domains. The paper also provides a clear mathematical formulation of the proposed method, which is a positive aspect for reproducibility. The paper also identifies the limitations of existing approaches that focus on next-turn rewards and attempts to address this by estimating a conversation-level reward. Finally, the paper's focus on long-term multi-turn interactions is a relevant and important area of research in the context of human-LLM collaboration.

Weaknesses

My analysis reveals several significant weaknesses in this paper, primarily concerning its presentation, methodological clarity, and experimental rigor. Firstly, the paper suffers from a lack of accessibility, as it immediately dives into technical details without providing sufficient context or background information. The introduction, for example, starts with equations and specific parameters without first establishing the broader context of the problem. Terms like 'ensemble-based reward estimation,' 'Bayesian meta-calibration,' and 'Monte Carlo sample counts' are introduced without sufficient explanation, making it difficult for a general audience to grasp the core concepts. This issue is further compounded in the 'main_idea' and 'method' sections, which also start with equations and technical jargon, assuming prior knowledge of multi-turn reward estimation and Bayesian methods. The term 'Bayesian meta-calibration' is used without a clear definition, and the explanation of how uncertainty is quantified is vague. This lack of clarity makes it challenging to understand the core contributions of the paper. My confidence in this assessment is high, as it is directly supported by the paper's content. Secondly, the paper lacks a clear, high-level overview of the proposed method. The 'method' section jumps directly into technical details without explaining the overall workflow or the interaction between the ensemble predictors, Bayesian meta-calibrator, and the clarification module. A diagram or a more detailed explanation of the information flow would greatly improve the reader's understanding. This lack of a clear overview makes it difficult to follow the proposed approach. My confidence in this assessment is high, as it is directly supported by the structure of the 'method' section. Thirdly, the paper's experimental evaluation is not sufficiently detailed. The 'experiments' section presents results without sufficient context, making it hard to understand the significance of the reported metrics. The 'results' section also suffers from a lack of detailed analysis, failing to explain the reasons behind the observed trends. The paper also lacks a clear explanation of how the synthetic data is generated and validated, and the experimental setup lacks crucial details, such as the specific LLMs used, the exact implementation of the ensemble predictors, and the hyperparameter tuning process. The paper also does not include a direct comparison with CollabLLM, a method mentioned in the related work, and the ablation studies do not provide sufficient insights into the contribution of individual components. The paper also does not discuss the computational overhead of the proposed method, which is an important consideration for practical applications. Finally, the paper lacks a dedicated 'Limitations' section, and the discussion of future work is brief and lacks specific details. The paper also does not provide a clear explanation of how the intrinsic reward is calculated, and the connection between the stated goal of balancing immediate and long-term effectiveness and the intrinsic reward's focus on token efficiency is not clear. The paper also does not provide a clear explanation of how the interactivity score is calculated. The paper also does not provide a clear explanation of how the synthetic data is generated and validated. The paper also does not provide a clear explanation of how the clarification module is implemented. The paper also does not provide a clear explanation of how the bonus adjustment is implemented. My confidence in these assessments is high, as they are directly supported by the paper's content and the lack of specific details in the experimental setup and results analysis.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the paper needs a significant restructuring to improve its accessibility. The introduction should begin with a clear and concise explanation of the problem being addressed, focusing on the challenges of long-term, multi-turn interactions between humans and LLMs. It should then introduce the core concepts of the proposed method, such as ensemble-based reward estimation and uncertainty-guided clarification, in a more intuitive and less technical manner. For example, instead of immediately presenting equations, the authors could explain the idea of using multiple reward predictors to get a more robust estimate and how uncertainty can be used to trigger clarifying questions. The 'method' section should also be reorganized to provide a high-level overview of the entire framework before diving into the technical details of each component. This overview should clearly explain the flow of information, from the input dialogue to the final clarification decision, and how each component contributes to this process. A diagram illustrating the framework would be beneficial. Secondly, the paper needs to provide more detailed explanations of the technical components. The 'Bayesian meta-calibration' process should be explained in more detail, including the specific Bayesian model used and how it is applied to the ensemble of predictors. The paper should also clarify how the uncertainty is quantified and used to trigger the clarification module. The connection between the conversation-level reward and the intrinsic reward should be made more explicit, and the rationale behind the specific form of the intrinsic reward should be explained in more detail. The paper should also provide a clear explanation of how the interactivity score is calculated. Thirdly, the paper needs a more rigorous experimental evaluation. The 'experiments' section should provide more context for the reported metrics, explaining why these metrics are appropriate for each task and how they relate to the overall goal of improving long-term collaboration. The 'results' section should include a more detailed analysis of the results, explaining the reasons behind the observed trends and discussing the limitations of the proposed method. The paper should also include a direct comparison with CollabLLM, a method mentioned in the related work, and the ablation studies should be expanded to

provide more insights into the contribution of individual components. The paper should also include a discussion of the computational overhead of the proposed method, including the time and memory requirements for training and inference. Finally, the paper should include a dedicated 'Limitations' section, which discusses the limitations of the proposed method and suggests directions for future research. The paper should also provide more details about the synthetic data generation process, including the specific prompts used and the criteria for selecting the data. The paper should also provide more details about the implementation of the clarification module and the bonus adjustment. These changes would significantly improve the clarity, rigor, and overall quality of the paper.

 Questions

My analysis raises several key questions that I believe are crucial for a deeper understanding of the paper's methodology and findings. Firstly, I am curious about the specific details of the synthetic data generation process. How exactly are the synthetic dialogues created, and what criteria are used to ensure their quality and relevance to the target tasks? Secondly, I would like to understand the rationale behind the specific choice of the Bayesian linear regression model for meta-calibration. Why was this model chosen over other Bayesian models, and what are its specific advantages and limitations in this context? Thirdly, I am interested in the details of the clarification module. How exactly is the clarification triggered, and what kind of clarifying questions are generated? Is the clarification process automated, or does it involve human intervention? Fourthly, I would like to know more about the computational overhead of the proposed method. What are the time and memory requirements for training and inference, and how do they compare to other methods? Fifthly, I am curious about the choice of the intrinsic reward function. Why is the intrinsic reward solely based on token efficiency, and how does this align with the stated goal of balancing immediate and long-term conversational effectiveness? What is the rationale behind the specific form of the intrinsic reward, and how was the value of λ set to 0.01? Sixthly, I would like to understand how the interactivity score is calculated. What are the specific features used to measure interactivity, and how are they combined to produce the final score? Finally, I am curious about the generalizability of the proposed method. How well does it perform on tasks and datasets that are different from those used in the experiments? These questions target core methodological choices and assumptions, and addressing them would significantly enhance the paper's clarity and impact.

 Scores

Soundness: 2.25

Presentation: 1.75

Contribution: 2.0

Rating: 3.0

Keywords

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Insights

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